

VIANA DO CASTELO, Portugal

WMO: 085430

Lat: 41.695N

Lon: 8.830W

Elev: 59

StdP: 14.66

Time Zone: 0.00 (EUW)

Period: 94-06

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	32.2	34.6	26.4	20.4	43.2	29.7	23.8	41.9	17.9	56.3	15.8	53.4	0.7	50	0.388

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	92.2	70.1	88.3	69.2	84.4	68.0	72.0	87.3	70.5	84.1	69.0	80.7	4.7	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	100.7	75.4	66.2	96.7	74.1	65.1	93.2	72.7	35.6	86.9	34.3	84.6	33.1	80.8	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.4	13.0	11.1	DB	28.2	98.2	3.0	2.4	26.0	99.9	24.2	101.3	22.5	102.6	20.3	104.3	(4)
(5)			WB	27.5	76.0	3.4	2.5	25.0	77.7	23.0	79.2	21.1	80.6	18.7	82.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.4	49.9	50.7	55.2	56.3	60.8	67.7	69.2	69.4	66.9	61.2	54.4	50.8		
	DBStd	8.60	4.77	4.42	4.68	4.37	5.16	5.96	4.95	4.50	4.11	4.34	4.74	4.71		
	HDD50	179	62	42	11	3	0	0	0	0	0	0	13	48		
	HDD65	2556	467	401	305	265	155	35	11	5	23	131	319	439		
	CDD50	3616	60	61	171	191	334	530	596	600	507	348	144	74		
	CDD65	517	0	0	1	3	23	115	141	140	80	14	0	0		
	CDH74	5025	0	0	52	82	309	1296	1361	1110	698	117	0	0		
	CDH80	1934	0	0	6	18	100	559	565	430	238	18	0	0		
(14) Wind	WSAvg	3.7	3.5	3.6	4.1	4.9	4.3	3.8	3.9	3.4	2.9	3.7	3.4	3.2		
(15) Precipitation	PrecAvg	56.0	7.5	6.7	4.3	4.4	4.0	2.1	0.9	1.1	3.6	6.1	6.9	8.3		
	PrecMax	75.0	17.0	19.0	13.6	13.7	10.0	7.4	3.2	3.8	14.4	21.0	19.4	23.0		
	PrecMin	33.8	0.7	0.2	0.0	0.8	0.2	0.0	0.0	0.0	0.0	0.3	0.0	1.1		
	PrecStd	10.6	4.2	4.3	3.1	2.9	2.4	1.4	0.9	1.1	3.1	4.1	4.4	4.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.4	70.0	80.2	82.8	89.1	95.1	95.2	97.6	91.0	82.4	71.7	64.8		
		MCWB	55.0	57.5	61.3	63.3	68.5	70.2	70.4	70.6	68.6	66.2	60.0	57.0		
	2%	DB	61.9	65.8	73.6	75.0	81.9	91.0	90.7	90.5	86.7	77.0	67.0	61.7		
		MCWB	54.3	55.3	57.8	59.8	64.5	69.9	70.4	70.4	68.3	63.7	59.0	56.7		
	5%	DB	59.8	62.5	69.5	69.9	76.5	86.7	86.7	84.2	81.9	72.8	64.5	60.2		
		MCWB	54.2	53.5	57.0	57.9	63.3	68.0	69.5	68.8	66.4	63.1	58.5	56.2		
	10%	DB	58.1	59.4	64.7	65.8	71.2	81.7	81.7	79.4	77.1	69.3	62.7	58.9		
		MCWB	54.4	52.6	55.3	56.5	61.0	66.8	68.0	67.2	65.3	62.3	58.0	55.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.2	59.7	62.4	65.1	72.4	72.9	73.8	74.8	71.0	68.4	64.1	60.4		
		MCDB	60.8	67.3	73.7	79.0	85.5	89.5	89.1	90.9	85.8	74.9	65.8	61.4		
	2%	WB	57.5	57.5	60.0	61.4	66.5	71.3	71.6	71.6	69.2	66.6	62.1	59.1		
		MCDB	58.4	62.1	67.7	71.5	78.8	88.2	86.9	84.6	81.2	72.3	64.3	60.2		
	5%	WB	56.6	55.9	58.6	59.4	64.1	69.3	70.0	69.9	67.8	65.0	60.5	57.9		
		MCDB	58.0	59.2	64.4	67.0	73.6	82.5	84.5	80.7	77.9	69.9	63.1	59.4		
	10%	WB	55.3	54.3	57.5	57.7	61.9	67.5	68.2	68.4	66.5	63.5	58.7	56.3		
		MCDB	57.9	58.0	62.7	63.2	69.0	78.5	79.8	77.5	74.9	67.8	61.5	58.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.5	18.2	18.1	16.8	17.4	20.0	20.0	19.8	19.9	16.4	15.8	15.2		
		MCDBR	18.4	24.6	27.6	27.0	27.7	30.2	29.7	28.7	29.3	23.5	19.2	16.5		
	5% WB	MCWBR	12.3	14.9	14.8	14.5	14.0	13.5	13.4	13.0	13.2	12.9	12.1	11.7		
		MCWBR	10.8	17.2	20.4	22.3	23.5	27.4	26.9	24.4	23.0	17.7	12.8	9.7		
(40) Clear-Sky Solar Irradiance	taub		0.340	0.348	0.383	0.382	0.388	0.391	0.380	0.392	0.380	0.374	0.347	0.335		
	taud		2.476	2.442	2.349	2.368	2.376	2.386	2.410	2.376	2.425	2.451	2.481	2.482		
	Ebn at Noon		255	272	273	282	282	280	282	275	271	258	250	246		
	Edh at Noon		24	29	36	37	38	38	36	36	32	28	24	22		
(44) All-Sky Solar Radiation	RadAvg		557	883	1236	1653	2004	2181	2207	1995	1557	996	644	503		
	RadStd		93	124	185	176	170	145	123	118	107	121	108	76		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+65

Nomenclature: See separate page